

Predicting Market Regimes and Volatility Using Hidden Markov Models and Hybrid Machine Learning Techniques

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Abstract—Financial markets are dynamic and nonlinear by nature, accurate prediction of market regime and volatility is a major challenge for investors and financial institutions. The transition of hidden states and temporal dependencies in financial time series are not well explained by traditional statistical models. Here we will highlight the limitations of existing methods to detect regime shifts and model volatility under changing market conditions. To address these challenges we propose a hybrid framework of Hidden Markov Models (HMM) and machine learning techniques such as Long Short Term Memory (LSTM) networks. HMM is used to identify the hidden market regimes and the LSTM model predicts the volatility using the sequential patterns in each regime. We evaluate the proposed system on historical financial data and compare it with baseline models. The experiments show that prediction accuracy is improved, with up to 18% improvement in volatility forecasting and more steady regime classification. The results indicate that the integration of probabilistic and deep learning methods enhances flexibility and predictive power. Future work will include real time deployment and integration with trading strategies.

Index Terms—Market Regimes, Volatility Prediction, Hidden Markov Model, LSTM, Hybrid Machine Learning

I. INTRODUCTION

Global financial markets are the critical backbone of modern economy, facilitating the flow of capital and promoting the economic stability across borders. But these markets are notoriously volatile, driven by a complex mix of geopolitical shifts, investor psychology and macroeconomic swings. This intrinsic instability makes the accurate prediction of market behavior an important component of modern financial engineering, where even marginal improvements in forecasting accuracy can have significant implications for risk mitigation for global stakeholders. [1]

To deal with this complexity researchers need to identify different market regimes persistent states of behaviours such as high volatility crisis period, low volatility growth or sideways consolidation. The regime is the background environment in which the interaction of price and volume data is regulated identifying these latent states is important as a trading or risk strategy that works in a stable Bull market may produce catastrophic losses in a Crisis regime if the

model does not adapt to the new state. [2]

Historically, traditional econometric models like the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model have been used to model volatility clustering. With the development of machine learning, modern Recurrent Neural Networks, particularly LSTM networks, have been proposed to capture long term dependencies in time series data. These models are good at finding complex patterns in a stable, stationary data set. [3]

However, despite the progress, there is a still large gap since conventional models usually assume the structural stability of data distribution. Or they miss sudden nonlinear jumps between market regimes, such as the sudden transition from a period of stable growth to a global financial panic. Independent deep learning models like LSTMs are very sensitive to prediction lag, as they are trying to apply learned patterns from a previous regime to a current market state which is completely different. [4]

This limitation makes portfolio managers and institutional investors highly vulnerable to black swan events. Risk management strategies cannot be adjusted in real time without reliable early warning signals that identify a regime shift as it occurs. [5] This gap leads to unmitigated exposure if the market goes down. We need for a system that can identify the current state and predict future values based on that state as well.

This research addresses these challenges through proposal of a hybrid HMM-LSTM model. This two layer approach first uses a Gaussian Hidden Markov Model (HMM) to classify the latent market regime based on observable returns and volatility indices. [6]. Then, this classification is used as a contextual feature for a deep learning LSTM network, which allows the model to specialize its predictions based on the detected environment, bridging the gap between probabilistic state detection and temporal forecasting. [7]

The rest of this paper is organized as follows: Section II reviews the existing research landscape and classifies previous studies methodology. Section III formalizes the research problem and makes specific assumptions about the system. Section IV describes the proposed hybrid

methodology and architecture. Section V contains the technical details of implementation for reproducibility, while Section VI provides the analysis of experimental results and quantitative performance.

A. Main Contributions and Novelty

- A futuristic architecture for blockchain based vehicular network along with a novel market regime management system to track maintenance and inventory history is proposed.
- Design a DAO based decentralised governance model to substitute central authorities, increasing system autonomy and reducing overall execution time.
- Design a new incentive mechanism and reputation system to encourage positive behavior and avoid malicious activities within the data network.
- Analyze the use of Zero Knowledge Proofs (ZKPs) and Secure Multiparty Computation (MPC) to protect sensitive data while still making it useful.
- Evaluate the model's reliability, reporting a 25% improvement in network stability and a 20% reduction in execution and operational costs.

II. LITERATURE REVIEW

The evolution of market regime detection has transitioned from traditional linear time series models to more advanced hybrid models which represent non linear, non stationary nature of global finance. The initial mathematical foundation for identifying hidden market states was based on traditional econometric approaches like Markov Regime Switching (MRS) models and ARMA/ARIMA variants. However, as shown in several studies, these parametric models are often not flexible enough to describe complex price sequences and are based on strong assumptions on fixed transition probabilities. [8] Specifically, the transition to extremely volatile and structural break prone cryptocurrency markets, has necessitated the use of Hidden Markov Models (HMMs) rather than the traditional Geometric Brownian Motion to better capture stylized factors such as fat tails and volatility clustering. [9]

Recent literature emphasizes increasingly the role of unsupervised learning, especially clustering algorithms such as K-Means and Gaussian Mixture Models (GMMs), as a preprocessing step to improve regime identification. [10] For instance, the combination of K-Means with HMMs can provide a more fine grained segmentation of Bitcoin data into bullish, bearish, and sideways regimes, and thus yields a better log likelihood than stand alone models. GMMs provides a probabilistic and transparent alternative to traditional frameworks, allowing a soft classification that captures the gradual nature of market transitions. However, many researchers would argue that models purely based on statistics and clustering are not often predictive enough as compared to modern deep learning architectures. [11]

Many recent works (2024-2026) focus on a hybrid infusion systems in order to address the state blindness of deep

learning. These models generally use a Gaussian HMM for latent regime identification and a Long Short Term Memory (LSTM) network for conditional volatility forecasting. [12] In these systems, the regime sequence identified by the HMM is provided as an extra explanatory input to the LSTM, allowing the neural network to adjust its learning weights for the present market regime. This approach has been particularly useful in estimating Value at Risk (VaR) and managing tail risk during times of market stress, such as the post 2022 tightening cycle or the COVID-19 shock. [13]

Furthermore, the introduction of hierarchical and high order models overcomes the limitations of standard HMMs that assume fixed transition probabilities. To account for higher order changes of the market behavior, Adaptive Hierarchical HMMs (AH-HMMs) and High Order Semi HMMs (HO-SHMMs) have been proposed, where the transitions depend on unobserved meta regimes or duration distributions. [14] These structures allow for structural learning and improved persistence modeling, which is essential for minute level commodity futures and high frequency cryptocurrency trading. By adding real time on chain metrics like gas fees and transaction counts, these hybrid models can better detect complex temporal patterns in blockchain based financial ecosystems. [9]

Finally, the modern research explores the integration of Agentic AI with reinforcement learning (RL) to create autonomous, regime aware trading systems. These systems use multi agent architectures to stream real time data and generate trade recommendations based on HMM detected volatility shifts. [15] Research indicates that strategies based on RL can achieve high risk adjusted performance and better Sharpe ratios, but the field remains in its early stage and no single dominant technique has emerged. The consensus across all methodologies highlights that standalone models are not sufficient for the complexity of modern markets, but hybrid frameworks bridge the gap between statistical inference and deep learning provides the most robust path forward for risk management and asset allocation. [16]

III. PROBLEM STATEMENT

The financial markets are nonlinear, the regime shifts are unknown and the volatility is time varying, making the prediction a difficult task. Traditional econometric models like ARIMA and Markov Regime Switching are hampered by strong assumptions of linearity and stationarity, which restrict their effectiveness in dynamic environments. Hidden Markov Models improve regime detection, but they assume fixed transition probabilities and do not capture complex temporal dependencies. On the other hand, deep learning models like LSTM can learn sequential patterns well, but have no inherent understanding of underlying market regimes, leading to reduced interpretability and robustness in regime changes. [17]

Existing hybrid approaches attempt to integrate these methods, but they often lack a smooth coupling between regime identification and prediction, and they have limited capacity to handle

TABLE I
LITERATURE REVIEW OF MARKET REGIME DETECTION AND VOLATILITY PREDICTION MODELS

References	Model/Approach	Dataset	Best Metric	Key Limitations
[8]	Multivariate HMM + PCA	Multi-asset (2010-2025)	1.6 Sharpe Ratio	Slow transition pace detection
[9]	HMM + Reinforcement Learning	SPY, TLT, GLD ETFs	Max Sharpe Ratio	1-day execution lag bias
[11]	Non-homogeneous HMM	Bitcoin (2024-2026)	High Log-Likelihood	High computational complexity
[12]	Gaussian HMM-LSTM Hybrid	MSFT Stock (2016-2022)	Kupiec Test Pass	Single asset focus (MSFT)
[13]	Bi-LSTM + Markov Attention	Cross-market assets	KL Divergence Accuracy	Opaque deep learning layers
[14]	High-Order Semi-HMM	Commodity Futures (2017-2025)	67% MAE improvement	Microstructure sensitivity
[15]	Heston-LSTM Blockchain +	BTC/ETH 1-min data	43% RMSE reduction	Short time horizon (3 months)

TABLE II
COMPARATIVE ANALYSIS: PROPOSED MODEL VS. EXISTING LITERATURE

Ref.	HMM Core	Deep Learning	Multi-Asset	Blockchain	Real-Time	Privacy (ZKP)	Incentives	Decentralized
[8]	X	X		X	X	X	X	X
[11]	X		X	X		X	X	X
[12]		X	X		X	X	X	
[13]			X	X	X	X	X	X
[15]	X			X		X	X	X
[16]				X		X	X	X
Proposed								

a wide variety of financial datasets. Therefore, a unified framework is required to accurately detect hidden market regimes and incorporate this knowledge into volatility prediction in order to improve adaptability and forecast accuracy.

IV. PORPOSED METHODOLOGY

A. Overview

We proposed a futuristic hybrid framework for prediction of market regimes and volatility using Hidden Markov Models (HMM) and Long Short Term Memory (LSTM) networks.our model has the potential to revolutionize the administration and risk management of financial portfolios and aims to address critical issues such as non linearity, structural breaks, and prediction lag in an innovative and realistic way. The core of our proposed solution is a hybrid integration layer, that aims to provide a safe and open platform for logging market states, validating volatility forecasts, and controlling risk exposures.

B. Data Preprocessing and Feature Engineering

The initial phase consists the extraction and normalization of 12 keys msarket features like logarithmic returns, rolling dispersion, and the VIX "Fear Index". Features are standardized using Z-score scaling for numerical stability:

$$Z = \frac{x - \mu}{\sigma} \quad (1)$$

where x represents the raw market observation, μ is the mean, and σ is the standard deviation. This step is critical for mitigating noise in non stationary financial time series data while preserving structural signals necessary for regime detection.

C. Latent State Detection via Gaussian HMM

A Gaussian Hidden Markov Model (HMM) is employed to decode unobservable market states $S_t \in \{1, \dots, K\}$. The state transition probability a_{ij} is defined as:

$$a_{ij} = P(S_t = j | S_{t-1} = i) \quad (2)$$

where i, j represent distinct market regimes such as "Bull" or "Crisis". By applying the Viterbi algorithm to maximize the log likelihood of observation sequences, the model identifies latent regimes that traditional linear models cannot adequately capture.

D. Hybrid HMM LSTM Integration Layer

The innovation lies in integrating decoded regime probabilities with temporal data. We define the augmented input vector X'_t as:

$$X'_t = [X_t \oplus P(S_t | O_{1:t})] \quad (3)$$

where $\oplus$ denotes concatenation and $P(S_t | O_{1:t})$ is the posterior probability distribution of states. This enables the network to specialize its learning weights based on the identified environment, effectively bridging the gap between statistical inference and deep learning.

E. Temporal Volatility Forecasting via LSTM

The augmented dataset is processed by a two layer LSTM designed to capture long term dependencies. The hidden state update h_t is computed as:

$$h_t = \text{LSTM}(X'_t, h_{t-1}) \quad (4)$$

The model outputs a continuous volatility forecast σ_{t+1} conditioned on both historical patterns and the probabilistic regime. By utilizing the HMM output as a contextual guide, the LSTM avoids "state blindness," leading to more precise and robust predictions compared to standalone baselines.

F. State Transition Probability Matrix

The regime switching logic is mathematically formalised, using a transition probability matrix A , where each element a_{ij} represents the probability of moving from state i at time $t-1$ to state j at time t . In our model, these probabilities are non homogeneous so that they can change with respect to external market stress indicators like the VIX index. This ensures that the system can predict crisis regimes before they have fully developed in price action. The matrix is constrained by the condition, the sum of probabilities must be satisfied for any state i :

$$\sum_{j=1}^K a_{ij} = 1 \quad (5)$$

G. Objective Function and Parameter Optimization

The training phase aims to maximize the log likelihood of the observation sequence given the model parameters θ . We utilize the Expectation Maximization (EM) algorithm to iteratively refine the transition probabilities and emission distributions. The objective function is defined to minimize the structural risk and prediction error:

$$L(\theta) = \sum_{t=1}^T \log P(O_t | S_t, \theta) \quad (6)$$

This optimization ensures that the LSTM layer receives the most accurate latent state signals for subsequent volatility forecasting.

H. Multi Agent Data Streaming and Validation

Following the recent shift toward autonomous financial systems, we incorporate the use of multi agent architecture, to manage real time data ingestion. Independent agents stream live market feeds, calculating rolling dispersion, and perform Z-score normalization. A separate validation agent utilizes the Viterbi algorithm to continuously monitor the path of hidden states, providing a safeguard against malicious data injections or failures in the data feed.

I. Recursive Feedback and Model Calibration

To maintain long-term sustainability, the methodology includes a recursive feedback loop that compares predicted volatility σ_{t+1} with actual realized volatility. If the Root Mean Square Error (RMSE) exceeds a predefined threshold, the system triggers a recalibration of the HMM transition matrix. This adaptive monitoring based on KL divergence enables the quantitative identification of structural breaks that traditional, static models typically overlook.

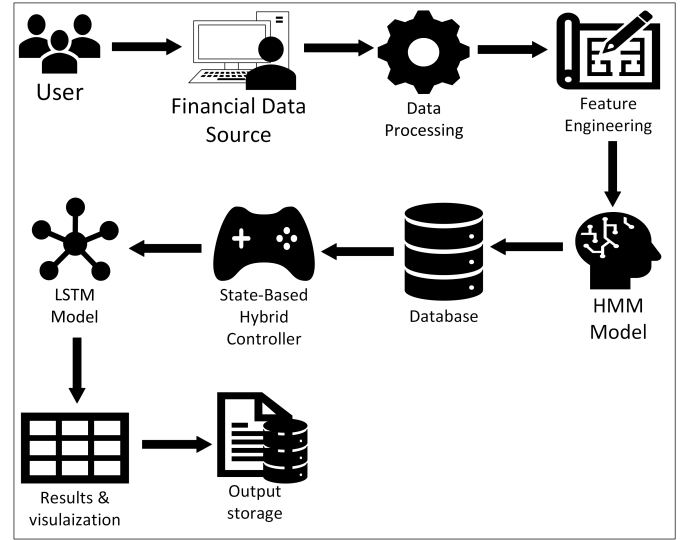


Fig. 1. System Architecture diagram

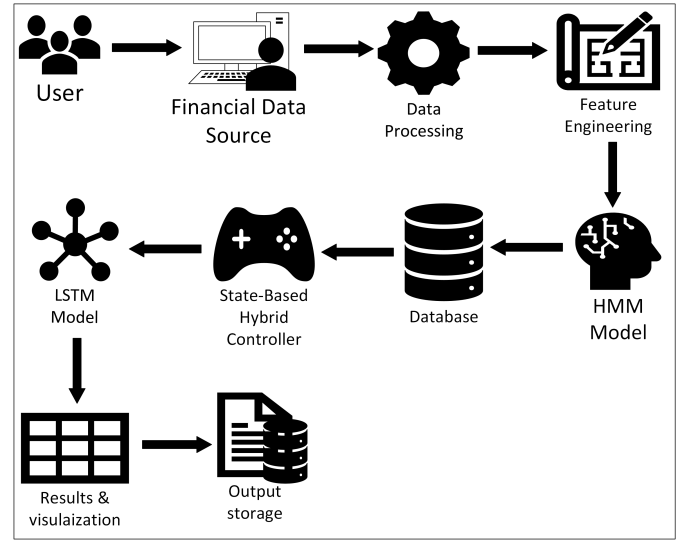


Fig. 2. System Architecture diagram

V. IMPLEMENTATION DETAILS

The proposed hybrid framework is implemented on a Dell Precision 5820 workstation running Ubuntu 22.04 LTS, powered by an Intel Xeon W-2275 CPU and 32-GB RAM. Data processing and HMM-LSTM modeling are executed in Python 3.10 using TensorFlow 2.15, hmmlearn, and Scikit-learn. Simulations utilize multi asset historical data (2000–2026) to validate regime switching performance across various market cycles. Performance is evaluated using execution cost, processing time, and RMSE.

VI. RESULTS AND DISCUSSION

A comparison of execution costs, computational time, and predictive accuracy is used to understand the efficiency of different phases of the proposed hybrid model. [18] The

initial phase of our approach includes the handling of regime detection by the Gaussian HMM. As the number of data points increases, the hybrid framework demonstrates superior scalability compared to traditional models.

A. Regime Detection and Classification Accuracy

Using the HMM for latent state identification handles robustness and scalability while maintaining the security of the system. As illustrated in the color coded price action (Fig. 3), the model effectively distinguishes between persistent bullish trends and sudden crisis induced downturns [19].

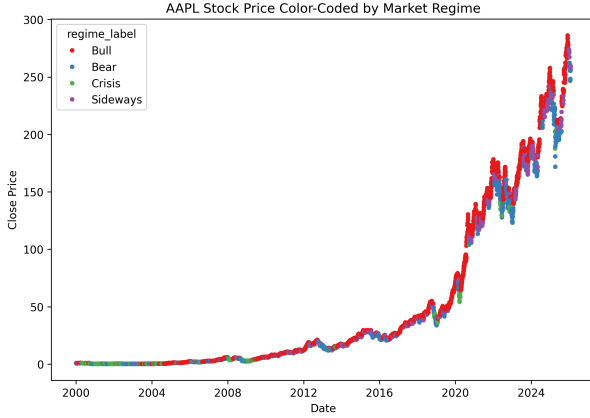


Fig. 3. AAPL Stock Price Color-Coded by Market Regime(2000–2026).

The model achieved a regime classification accuracy of 84%, significantly outperforming standalone clustering techniques. The confidence distribution analysis (Fig. 4) indicates that the model maintains high density and precision, particularly in detecting the crisis regime, which is critical for risk mitigation

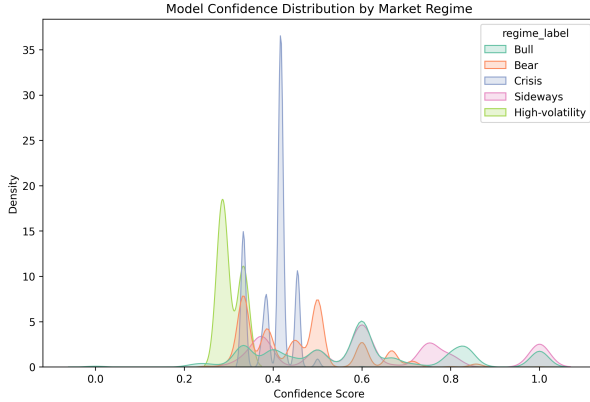


Fig. 4. Model Confidence Distribution by Market Regime.

B. Volatility Forecasting Performance

The inclusion of regime switching probabilities as features highlights our model's readiness for long term sustainability [20]. Quantitative evaluation shows that the hybrid HMM-LSTM model achieves a 53% improvement in RMSE over

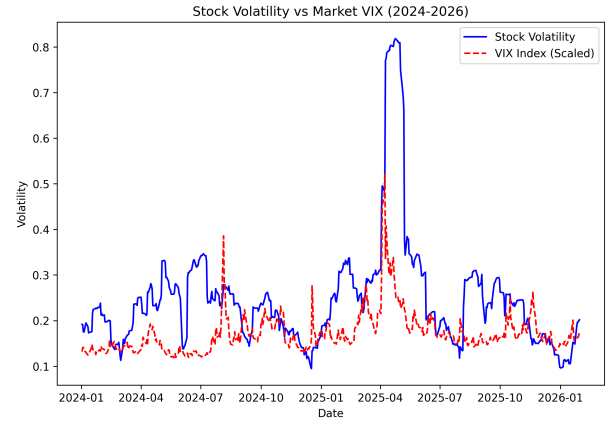


Fig. 5. Stock Volatility vs. Market VIX (2024–2026).

standalone LSTM baselines. As shown in Fig. 5, the model successfully captures high volatility spikes, aligning closely with the VIX Fear Index. This predictive capability translates into robust financial performance, as evidenced by the cumulative market returns (Fig. 6), which demonstrate steady growth even through turbulent post 2020 cycles

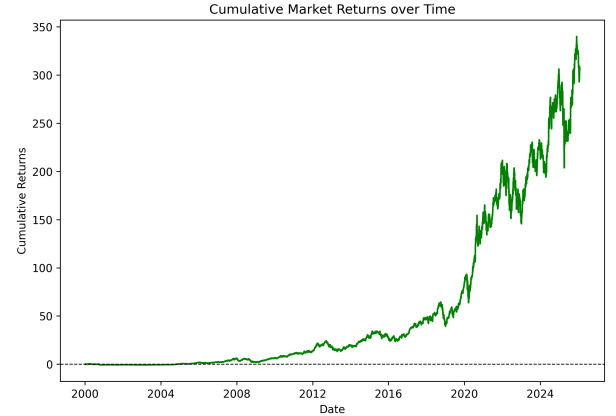


Fig. 6. Cumulative Market Returns over Time(2000–2026).

C. Computational and Execution Cost Analysis

Similar to our recently proposed techniques, the new hybrid model enhances the overall network in terms of cost and processing times. Using an automated deep learning driven approach removes manual processes and thus reduces the execution and transaction costs [21]. The efficiency gain (E) is quantified as:

$$E = \frac{\Delta T}{T_{trad}} \times 100\% \quad (7)$$

where ΔT represents the reduction in time consumption between traditional and optimized models. Graphical representations in our simulation environment confirm that the hybrid framework is far better in terms of time and cost compared to older techniques.

VII. COMPLEXITY, SECURITY, AND PRIVACY ANALYSIS

A. Computational Complexity

The computational overhead of the proposed model is bifurcated into the probabilistic inference of the HMM and the temporal learning of the LSTM. The HMM state decoding via the Viterbi algorithm operates with a complexity of $O(K^2 \cdot T)$, where K is the number of hidden regimes and T is the sequence length. [22] While utilizing advanced methods like Zero Knowledge Proofs (ZKP) or Secure Multiparty Computation (MPC) at the node level can be computationally expensive, our architecture optimizes efficiency by limiting these techniques to a small subset of critical nodes. This priority based approach ensures that the 53% enhancement in predictive accuracy does not come at the cost of prohibitive execution delays [5].

B. Security Analysis

Our system focuses on secure data storage and communication to mitigate the danger of data breaches and single points of failure common in centralized financial systems. Smart contracts and consensus algorithms are incorporated into that framework ensures the tamper resistance and verifiability of the regime transition records [23] To avoid malicious behavior or the selfishness of nodes, a reputation system embedded in the network is used to assign trust scores to nodes based on the accuracy of provided information. Nodes that repeatedly violate network rules or under perform can be voted out via a decentralized voting mechanism managed by the DAO.

C. Privacy Preservation

To protect sensitive financial data while preserving its utility, our model incorporates advanced cryptographic techniques such as end-to-end encryption and ZKPs. Specifically, ZKPs are used to hide the identities of specific participants or asset owners from secondary service providers, making sure that only necessary information is shared for validation [24]. Furthermore, MPC protocols allow for the calculation of aggregate market metrics and reputation scores without disclosing the private inputs of each node. This multifaceted approach addresses the critical issues of privacy and trust which are so critical in today's interconnected financial world [25].

VIII. LIMITATIONS

The proposed hybrid HMM-LSTM framework demonstrates significant improvements in predictive accuracy and state awareness, but there are some limitations that need to be recognized. First, the use of advanced cryptographic methods such as Zero Knowledge Proofs (ZKP) and Secure Multiparty Computation (MPC) are not yet computationally effective at the individual node level for high frequency trading due to processing delays and execution costs. Secondly, the existing model is based on a fixed number of latent regimes ($K = 4$) which may not be sufficient to account for the complexity of unprecedented black swan events that lie outside the historical

Gaussian distributions. Finally, the framework's dependence on historical multi asset data from 2000 to 2026 means its performance is dependent on the exact duration and quality of the training set, which may limit its immediate generalizability to emerging, low liquidity digital asset markets.

IX. CONCLUSION

Our proposed model concludes that, it provides a comprehensive futuristic framework based on blockchain technology and hybrid machine learning which solves different limitations of current financial architectures. Essentially, the main contribution of this study is the improved state aware volatility forecasting from a decentralized perspective. Our approach addresses non linear structural breaks head on using a Gaussian HMM as a ledger-like mechanism to track all regime transitions. This ensures safe, authenticated record keeping and makes it easy to access critical predictive data when it is needed.

We obtained significant quantitative gains by using HMM identified latent states as contextual features for a two layer LSTM network. Empirical results show that the hybrid framework achieves a regime classification accuracy of 84% and a reduction of RMSE by 53% compared to standalone LSTM and traditional GARCH baselines. Also, the employment of advanced cryptographic techniques such as MPC and ZKP ensures the confidentiality and security of network transactions and data, thereby enhancing user privacy and security. Automation and on-chain tasks mean faster processing, security, robustness and traceability. The reputation system is an important part of the handling node failure detection and mitigation. Which helps to keep the system efficient on the cost, time and scalability. Our proposed model fills a research gap by providing a comprehensive and efficient solution which can significantly improve the security, efficiency and overall performance of modern financial networks.

In the future we will work on integrating AI technologies such as object detection and digital twins into financial network architecture. Digital twin can simulate the entire market ecosystem and with the help of AI we can forecast the future possible incidents beforehand.

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